Neurological manifestations of lysosomal disorders and emerging therapies targeting the CNS

Roberto Giugliani, Filippo Vairo, Francyne Kubaski, Fabiano Poswar, Mariluce Riegel, Guilherme Baldo, Jonas Alex Saute

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Medical Genetics Service, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil (Prof R Giugliani MD, F Poswar MD, M Riegel PhD, J A Saute MD)**; Department of Genetics** (Prof R Giugliani), **Postgraduate Program in Genetics and Molecular Biology** (Prof R Giugliani, F Poswar, M Riegel, G Baldo PhD), **Postgraduate Program in Physiology** (G Baldo), **and Postgraduate Program in Medicine: Medical Sciences** (J A Saute), **Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil; Mayo Clinic, Rochester, MN, USA** (F Vairo MD); **and University of Delaware, Newark, DE, USA** (F Kubaski PhD)

Lysosomal disorders have been an area of interest since intravenous enzyme replacement therapy was successfully introduced for the treatment of Gaucher's disease in the early 1990s. This treatment approach has also been developed for several other lysosomal disorders, including Fabry's disease, Pompe's disease, lysosomal acid lipase deficiency, and five types of mucopolysaccharidosis. Despite the benefits of enzyme replacement therapy, it has limitations—most importantly, its ineffectiveness in treating the neurological components of lysosomal disorders, as only a small proportion of recombinant enzymes can cross the blood–brain barrier. Development of strategies to improve drug delivery to the CNS is now the primary focus in lysosomal disorder research. This Review discusses the neurological manifestations and emerging therapies for the CNS component of these diseases. The therapies in development (which are now in phase 1 or phase 2 clinical trials) might be for specific lysosomal disorders (enzyme replacement therapy via intrathecal or intracerebroventricular routes or with fusion proteins, or gene therapy) or applicable to more than one lysosomal disorder (haemopoietic stem cell transplantation, pharmacological chaperones, substrate reduction therapy, or stop codon readthrough). The combination of early diagnosis with effective therapies should change the outlook for patients with lysosomal disorders with neurological involvement in the next 5–10 years.

Introduction

Lysosomal disorders are mainly caused by mutations in genes encoding lysosomal hydrolases, which are responsible for the degradation of macromolecules inside cells.[1] Defects in the post-translational processing of lysosomal enzymes, enzyme trafficking and targeting, enzyme cofactors, and the function of non-enzymatic lysosomal transmembrane and soluble proteins also cause cellular disturbances, leading to lysosomal disorders.[2]

Classification of lysosomal disorders is generally based on either the biochemical nature of the storage material or the underlying disease-causing mechanism.[1] The former classification, although clinically useful, does not take into account that several different substrates are stored in many lysosomal disorders, whereas the latter is more suitable for the study of disease pathogenesis and design of new therapies. Table 1 shows an adapted classification based on both models.[3–21]

Although lysosomal enzymes are products of housekeeping genes that are expressed ubiquitously, their substrates have a much less uniform distribution across cell types. Substrate expression, therefore, has a key role in the selectivity of tissues affected by storage.[1] In sphingolipidoses, for example, the cell can only store a specific glycosphingolipid if it synthesises it or acquires it from an external source. For instance, in type 1 Gaucher's disease, the lack of glucocerebrosidase activity leads to storage of glucosylceramide—which is acquired following phagocytosis of senescent leucocytes—in macrophages. In lysosomal disorders that affect the CNS, a similar process, such as apoptotic cell clearance, could lead to storage in microglial cells, whereas storage in neurons arises because they synthesise the lipid in question. The pattern of storage in the CNS is cell-type dependent, as different neurons have different glycosphingolipid expression profiles, and pathology is therefore confined to certain cell populations.

Differences in cellular vulnerability to substrate accumulation and to other cellular dysfunctions related to lysosomal disorders also determine which tissues are affected.[22] CNS involvement occurs in two-thirds of lysosomal disorders, despite a lower concentration of storage materials in the brain than in other organs.[2,22] Neuronal vulnerability is probably related to the lack of compensatory cellular metabolic pathways and to limited cell regeneration.[2] Storage in the CNS might also affect glial cells, resulting in abnormal formation or maintenance of myelin, astrocytosis, and activation of microglia, which can be key mediators of neurodegeneration.

Key messages

- Lysosomal disorders are multisystemic and progressive conditions, and many of them have neurological involvement, which often dominates disease presentation.
- Intravenous enzyme replacement therapy, which is available for several lysosomal disorders, does not address the neurological component of the disease, as the recombinant enzymes do not cross the blood–brain barrier effectively.
- Several novel therapeutic strategies (including intrathecal or intracerebroventricular delivery of enzymes, fusion proteins that cross the blood–brain barrier, substrate reduction therapy, pharmacological chaperones, and gene therapy) are being developed to treat the neurological component of many lysosomal disorders, and should become available soon.
- Knowledge about the natural history of the neurological manifestations of lysosomal disorders is important to define outcomes for clinical trials, and the identification of the minimally important differences will be paramount for assessment of long-term and short-term efficacies of new therapies for the different lysosomal disorders.
- The combination of early diagnosis with effective therapies should change the outlook for patients with lysosomal disorders with neurological involvement in the next 5–10 years.

Neurological manifestations

Lysosomal disorders are often multisystem disorders, although in some neuronopathic subtypes disease

expression is limited to the CNS.[4] Infants with predominant neuronopathic lysosomal disorders often appear normal at birth, but the age of disease onset can vary considerably among different lysosomal disorder subtypes and even within one subtype, ranging from severe infantile-onset to mild adult-onset forms. Developmental delay and regression of neurological function occur together with numerous neurological manifestations such as abnormal ocular movements, ataxia, seizures, extrapyramidal signs, spasticity, visual loss, and neurocognitive disorders that are usually progressive and untreatable.[22] The variation in the timing of disease onset is mostly related to residual activity of the metabolic pathway involved, because just a small amount of residual activity can prevent storage in most tissues. However, correlations between genotype and phenotype are not always well defined.[22] Table 1 depicts age at onset and the most frequent neurological signs of the different forms of lysosomal disorders, considering both primary and secondary neurological involvement and predominance of involvement of the CNS or peripheral nervous system.[3–21]

Cellular biology of neuronopathic forms

In the past, signs and symptoms of neuronopathic lysosomal disorders were considered to be related to the mechanical disruption of neurons resulting from storage, leading to cellular dysfunction and death. However, evidence from the past two decades has shown that the cellular scenario is much more complex than would be predicted by a simple physical disruption model and that neuronal dysfunction might even occur independently from storage burden.[4] Furthermore, lysosomal storage often does not consist of a one enzyme–one substrate relationship, but rather involves concomitant and different storage products. Not only do multiple storage products result from a lack of hydrolase substrate specificity, but often they also represent secondary storage that is unrelated to the primary protein defect. For example, GM2 and GM3 gangliosides are secondarily accumulated

Correspondence to:
Prof Roberto Giugliani, Medical Genetics Service, Hospital de Clínicas de Porto Alegre, Porto Alegre, RS 90035-903, Brazil
rgiugliani@hcpa.edu.br

	Deficiency	Storage	Onset	CNS involvement	PNS involvement	Other signs
Lysosomal hydrolase defect						
Gaucher's disease (type 2)[3,4]	Glucosylceramidase	Glucosylceramide	Early infancy	Oculomotor palsies; cerebellar signs; UMN	None	Stridor and other bulbar signs; systemic involvement
Gaucher's disease (type 3)[3,4]	Glucosylceramidase	Glucosylceramide	Childhood or adolescence	Abnormal horizontal saccades; oculomotor apraxia; myoclonus; seizures; dementia (late stages); cerebellar signs; extrapyramidal findings; UMN	None	Systemic involvement
GM1 gangliosidosis[4,5]	β-Galactosidase	GM1 ganglioside	Early infancy to adulthood	CRS (usually absent in late-infantile and adult forms); seizures; dementia (late stages); extrapyramidal findings (late-onset forms); UMN	None	Systemic involvement; skeletal dysplasia; corneal clouding; exaggerated startle response; deafness; cardiomyopathy
Tay-Sachs disease[4,6,7]	β-Hexosaminidase A	GM2 ganglioside and related glycolipids	Early infancy to adulthood	CRS (usually absent in late-infantile and adult forms); optic atrophy and retinopathy (late infantile and juvenile); myoclonus (early infantile, myoclonic jerks); seizures; psychosis (adult form); dementia and cerebellar signs (juvenile and adult forms); extrapyramidal and UMN findings (adult form); exaggerated startle	LMN (adult form)	Macrocephaly (early infantile in late stages); systemic involvement
Sandhoff's disease[4,6,7]	β-Hexosaminidase A and β-hexosaminidase B	GM2 ganglioside and related glycolipids	Early infancy to adulthood	CRS (usually absent in late-infantile and adult forms); optic atrophy and retinopathy (late infantile and juvenile); myoclonus (early infantile, myoclonic jerks); seizures; psychosis (adult form); dementia and cerebellar signs (juvenile and adult forms); extrapyramidal and UMN findings (adult form); exaggerated startle	LMN (adult form)	Macrocephaly (early infantile in late stages); systemic involvement; skeletal dysplasia
Fabry's disease[8]	α-Galactosidase A	Globotriaosylceramide and galactosyl-ceramide	Childhood to adulthood	Depression; stroke with different neurological syndromes	PNP (autonomic involvement, prominent sensory involvement)	Angiokeratoma; renal failure; hypertrophic cardiomyopathy; hearing loss and tinnitus; cornea verticillata
Krabbe's disease[4,9]	β-Galactosyl-ceramidase	Psychosine	Early infancy to adulthood	Optic atrophy; behavioural symptoms, dementia and cerebellar signs (later-onset forms); UMN	PNP	Characteristic leukodystrophy
Niemann-Pick type A[4,6]	Acid sphingomyelinase	Sphingomyelin	Early infancy	CRS	PNP	Systemic involvement
Metachromatic leukodystrophy[4,10]	Arylsulphatase A	Galactosyl sulphatide; lactosyl sulphatide	Early infancy to adulthood	Optic atrophy and retinopathy (late-infantile form); CRS (rarely); seizures; behavioural symptoms, dementia, cerebellar signs, dystonia (later-onset forms); UMN	PNP	Characteristic leukodystrophy

(Table 1 continues on next page)

in affected neurons in Niemann-Pick disease type C and in different mucopolysaccharidosis subtypes.[2]

For most lysosomal disorders, pathology involves neuronal dysfunction rather than neuronal loss, with cellular death characterising the late stages of these diseases. Exceptions are the differential loss of Purkinje neurons that characterises some storage diseases, including Niemann-Pick disease type C, and the massive

	Deficiency	Storage	Onset	CNS involvement	PNS involvement	Other signs
(Continued from previous page)						
Mucopolysaccharidosis						
Type I[4,11]	α-Iduronidase	Dermatan sulphate; heparan sulphate	Early infancy	Optic atrophy and retinopathy (not frequent); dementia (Hurler); UMN (secondary to spinal cord compression)	Carpal tunnel syndrome	Coarse facies; skeletal dysplasia; systemic involvement; valvar cardiac involvement; hearing loss; corneal clouding
Type II[12]	Iduronate sulphatase	Dermatan sulphate; heparan sulphate	Early infancy	Optic atrophy and retinopathy; seizures; behavioural symptoms; dementia; UMN (secondary to spinal cord compression)	Carpal tunnel syndrome	Coarse facies; skeletal dysplasia; systemic involvement; valvar cardiac involvement; corneal clouding (occasional)
Type IIIA[6]	Heparan sulphamidase	Heparan sulphate	Childhood	Optic atrophy and retinopathy (not frequent); seizures; behavioural symptoms; dementia; cerebellar and UMN signs	LMN and PNP	Coarse facies (less prominent); hearing loss
Type IIIB[6]	α-*N*-acetyl-glucosaminidase	Heparan sulphate	Childhood	Optic atrophy and retinopathy (not frequent); seizures; behavioural symptoms; dementia; cerebellar and UMN signs	LMN and PNP	Coarse facies (less prominent); hearing loss
Type IIIC[6]	Acetyl CoA: α-glucosaminide *N*-acetyltransferase	Heparan sulphate	Childhood	Optic atrophy and retinopathy (not frequent); seizures; behavioural symptoms; dementia; cerebellar and UMN signs	LMN and PNP	Coarse facies (less prominent); hearing loss
Type IIID[6]	*N*-acetyl-glucosamine-6-sulphatase	Heparan sulphate	Childhood	Optic atrophy and retinopathy (not frequent); seizures; behavioural symptoms; dementia; cerebellar and UMN signs	LMN and PNP	Coarse facies (less prominent); hearing loss
Type VII[6]	β-Glucuronidase	Heparan sulphate; dermatan sulphate; chondroitin-4 and chondroitin-6 sulphates	Early infancy	Optic atrophy and retinopathy (not frequent); UMN (secondary to spinal cord compression)	Carpal tunnel syndrome	Coarse facies; skeletal dysplasia; systemic involvement; valvar cardiac involvement; hearing loss; corneal clouding
Farber's disease[4,6,13]	Acid ceramidase	Ceramide	Early infancy and childhood	Retinopathy; myoclonus in the spinal muscular atrophy with progressive myoclonic epilepsy phenotype; UMN	PNP; LMN (spinal muscular atrophy with progressive myoclonic epilepsy phenotype)	Joint deformities; lipomatous skin nodules
Pompe's disease[6,14]	α1,4-Glucosidase	Glycogen	Early infancy to adulthood	None	LMN; myopathy	Hypertrophic cardiomyopathy
α-Mannosidosis[4,15]	α-Mannosidase	Oligomannosides	Perinatal to adulthood	Behavioural symptoms; cerebellar and UMN signs (not frequent)	Myopathy	Coarse facies; skeletal abnormalities; systemic involvement
β-Mannosidosis[4,6,16]	β-Mannosidase	Oligosaccharides	Variable	Seizures; behavioural symptoms; cerebellar and UMN signs	PNP	Coarse facies; skeletal abnormalities; angiokeratoma
Schindler disease[4]	*N*-acetylgalactosa-minidase	*O*-linked glycopeptides	Early infancy	Optic atrophy and retinopathy; myoclonus; seizures; UMN	None	None
Sialidosis[4,6]	α-Neuraminidase	Sialic-acid-containing oligosaccharides	Congenital to adolescence	CRS; myoclonus; seizures; cerebellar and UMN signs (type 2)	None	Dysmorphic features (type 2); systemic involvement (type 2)
NCL1[17]	Palmitoyl-protein thioesterase 1	Lipidated thioesters	Infancy	Optic atrophy and retinopathy; myoclonus; seizures; dementia; cerebellar and UMN signs	None	None
NCL2[17]	Carboxypeptidase	Subunit *c* of the mitochondrial ATP synthase	Late infancy	Optic atrophy and retinopathy; myoclonus; seizures; dementia; cerebellar, extrapyramidal, and UMN signs	None	None

(Table 1 continues on next page)

	Deficiency	Storage	Onset	CNS involvement	PNS involvement	Other signs
(Continued from previous page)						
Post-translational processing defect in lysosomal enzymes						
MSD[4,6]	Multiple sulphatases	Sulphatides	Early infancy	CRS (not frequent); optic atrophy and retinopathy (late-stage forms); myoclonus; seizures; UMN	PNP	Facial and skeletal deformities; systemic involvement; leukodystrophy
Trafficking defect in lysosomal enzymes						
Mucolipidosis type II (also known as I-cell disease)[4,6,18]	*N*-acetylglucosamine phosphate transferase	Oligosaccharides; mucopolysaccharides; lipids	Early infancy	Intellectual disability	None	Coarse facies; skeletal deformities
Mucolipidosis type IIIA (pseudo-Hurler)[4,6,18]	*N*-acetylglucosamine phosphate transferase	Oligosaccharides; mucopolysaccharides; lipids	Early infancy	Intellectual disability	Carpal tunnel syndrome	Coarse facies; skeletal deformities
Defect in lysosomal enzyme protection						
Galactosialidosis[4,6]	Protective protein cathepsin A; β-galactosidase; neuraminidase	Sialyl-oligosaccharides	Early infancy	CRS; seizures; UMN	None	Coarse facies; systemic involvement
Defect in soluble non-enzymatic lysosomal proteins						
Niemann-Pick type C[19]	NPC2	Cholesterol; sphingolipids	Early infancy to adulthood	VSGP; seizures; behavioural symptoms; dementia; cerebellar, extrapyramidal, and UMN signs	None	Systemic involvement; neonatal jaundice; gelastic cataplexy; dysphagia
GM2 activator protein deficiency, variant AB[20]	GM2 activator protein	GM2 ganglioside and related glycolipids	Early infancy and adolescence	CRS; seizures; cerebellar and UMN signs; exaggerated startle	None	None
Transmembrane (non-enzyme) protein						
Danon disease[14]	LAMP2	Cytoplasmic debris; glycogen	Adolescence	None	Myopathy	Hypertrophic cardiomyopathy
Niemann-Pick type C[19]	NPC1	Cholesterol; sphingolipids	Early infancy to adulthood	VSGP; seizures; behavioural symptoms; dementia; cerebellar, extrapyramidal, and UMN signs	None	Systemic involvement; neonatal jaundice; gelastic cataplexy; dysphagia
Salla disease[6]	Sialin	Sialic acid	Infancy	Seizures (rare); intellectual disability; cerebellar, extrapyramidal, and UMN signs	None	Mild coarse features
NCL3[17]	CLN3	Subunit *c* of the mitochondrial ATP synthase	Adolescence	Optic atrophy and retinopathy; myoclonus; seizures; dementia; cerebellar, extrapyramidal, and UMN signs	None	None
Mucolipidosis type IV[6,21]	Mucolipin	Gangliosides; phospholipids; mucopolysaccharides	Early infancy	Optic atrophy and retinopathy; intellectual disability; UMN	None	None

CNS=central nervous system. PNS=peripheral nervous system. UMN=upper motor neuron. CRS=cherry red spot. LMN=lower motor neuron. PNP=peripheral polyneuropathy. NCL=neuronal ceroid lipofuscinosis. MSD=multiple sulphatase deficiency. VSGP=vertical supranuclear gaze palsy.

***Table 1*: Classification of lysosomal disorders according to molecular defect and main neurological findings**

cell loss that occurs in the neuronal ceroid lipofuscinoses.[2] Neurodegeneration in lysosomal disorders seems to result from a complex and still not fully understood cascade of events associated with secondary changes in neurons, altered axonal transport, dysfunction of intracellular protein degradation and macroautophagy (referred to as autophagy from here) pathways, mitochondrial dysfunction, altered calcium storage and endoplasmic reticulum functions, and inflammation.[2]

Neuropathology in neuronopathic forms

Meganeurites and axonal spheroids are the main morphological abnormalities observed in neurons in lysosomal disorders, except neuronal ceroid lipofuscinoses. Meganeurites are huge axon hillock enlargements containing storage material consistent with the specific defective lysosomal hydrolase. Their volume often exceeds that of adjacent perikarya, and they are associated with increased expression of GM2 ganglioside. GM2 has a major regulatory role in cortical pyramidal neuron dendritogenesis as a neurite growth-promoting factor or as a growth-factor acceptor.[2] In lysosomal disorders with ganglioside storage, cortical pyramidal neurons also exhibit abnormal sprouting of new dendrites at the axon hillock (called ectopic dendritogenesis). Asymmetrical synapses are known to occur on these dendrites, and this input is likely to be injurious to neuron function.[2]

Axonal spheroids are focal granular enlargements within axons (also known as neuroaxonal dystrophy) that

contain a distinctly different range of materials not related to the specific defective lysosomal hydrolase, and the ultrastructural features of spheroids, unlike those of meganeurites, are the same across many types of storage disorders.[2] In lysosomal disorders, spheroids are found in abundance in primary ganglioside storage diseases, Niemann-Pick disease (types A and C), and the glycoproteinosis α-mannosidosis.[2] Spheroids are found most commonly on GABAergic neurons (GABA is the main inhibitory neurotransmitter in the CNS), including Purkinje cells, neurons in the basal ganglia, and non-pyramidal (intrinsic) cells of the cerebral cortex, and their distribution correlates with the neurological manifestation of the disease. They can interfere with both axoplasmic anterograde and axoplasmic retrograde transport. Defects in inhibitory circuits related to spheroids, possibly as a result of timing or other transmission effects rather than outright cell death, might be an important factor underlying brain dysfunction in many storage diseases.[2]

By contrast with other primary neuronopathic lysosomal disorders, patients with neuronal ceroid lipofuscinoses have neuronal loss in the cerebral and cerebellar cortices and accumulation of autofluorescent material (so-called ceroid lipopigments) and subunit *c* of mitochondrial ATP synthase (except in type 1 disease) in the lysosomes of neurons and other cells in most tissues.[2]

Necroptosis, an unusual cell-death mechanism, has been described in three sphingolipid storage diseases: type 2 Gaucher's disease, Krabbe's disease (also known as globoid cell leukodystrophy), and Niemann-Pick disease type C. Necroptosis is a major caspase-independent programmed cell-death pathway that is dependent on the kinases RIPK1 and RIPK3. Expression of RIPK1 and RIPK3 was significantly elevated in a mouse model of neuronopathic Gaucher's disease and correlated with the extent of pathology in different brain regions.[23] Increased RIPK1 expression was also found in the post-mortem brain of a patient with type 2 Gaucher's disease.[23] The necroptosis pathway was activated in a Krabbe's disease mouse model, but not in models of other lysosomal disorders, including Sandhoff's disease and GM1 gangliosidosis. In a recent study,[24] increased expression of RIPK1 and RIPK3 was found in fibroblasts and neuronal precursor cells derived from induced pluripotent stem cells of patients with Niemann-Pick disease type C, and in cerebellar tissue from both transgenic mice and patients. Notably, pharmacological inhibition of RIPK1 decreased cell death in patient-derived fibroblast cell lines in vitro, and significantly delayed cerebellar Purkinje cell loss, progression of neurological symptoms, and death in the transgenic Niemann-Pick C1 mice in vivo.[24] Therefore, targeting of necroptosis opens up a new avenue for the therapy of these severe and still untreatable neuronopathic lysosomal disorders.

Neurodegenerative mechanisms in lysosomal disorders are similar, even though the ultimate cause and storage materials can vary greatly. Because of their structural and functional characteristics, neurons rely on active membrane transport and traffic of trophic factors through cell bodies, distant axons, and dendrites for survival. Therefore, neurons are very sensitive to the accumulation of damaged or aggregated cytosolic proteins, or membranes. Lysosomal storage impairs autophagy, resulting in the accumulation of polyubiquitinated proteins and dysfunctional mitochondria, which ultimately leads to apoptosis.[2] Impairment of autophagy and accumulation of autophagy substrates in lysosomal disorders suggest that at least some mechanisms underlying the disease phenotype might be similar to those of other diseases in which defective autophagy has been observed. In support of this idea, intraneuronal accumulations of proteins have been detected in many neurodegenerative diseases that affect the ageing population, including Alzheimer's disease, Parkinson's disease, and polyglutamine diseases.[14]

Neuromuscular involvement

Much attention has been given to CNS involvement in lysosomal disorders, but little to the involvement of peripheral neuromuscular systems, even though the neurological manifestations of some of these disorders are limited to this system (table 1). Both post-mitotic skeletal muscle and peripheral nerve cells[2] are susceptible to storage and to other cellular changes related to lysosomal disorders.

After years of preclinical and clinical experience with enzyme replacement therapy in Pompe's disease, it became clear that whereas cardiac muscle responds very well to such therapy, skeletal muscle does not.[14] Large areas of autophagic accumulation were detected in therapy-resistant fast muscle, which suggested that abnormal autophagy might be the reason for the lack of response. From a morphological perspective, abnormal autophagy, rather than lysosomal expansion, seems to be the major cause of muscle fibre destruction and the loss of muscle strength. Both the generation of autophagosomes and their fusion with, and degradation by, lysosomes are impaired in skeletal muscles of patients with Pompe's disease.[14] Danon disease is another lysosomal glycogen storage disease characterised by cardiomyopathy and variable skeletal muscle weakness in which accumulation of autophagic vacuoles is detected in several tissues, particularly the muscle.[14]

Fabry's disease, which is caused by α-galactosidase A deficiency, leads to storage of neutral glycosphingolipids, particularly globotriaosylceramide and galactosylceramide. Pain and acroparesthesia are thought to be caused by either poor perfusion of the peripheral nerves or lysosomal accumulation of glycosphingolipids in neurons, dorsal root ganglia, and the spinal cord, which leads to atrophy of the small unmyelinated nerves.[8]

Secondary neurological involvement

Neurological dysfunction in lysosomal disorders can also result from secondary complications of storage materials, such as spinal cord compression in mucopolysaccharidosis types I, IV, or VI, and stroke in Fabry's disease.[4] Spinal involvement in mucopolysaccharidosis can be a consequence of vertebral and skull base anomalies, glycosaminoglycan deposits in the meninges and supporting ligaments, as well as atlantoaxial subluxation and instability, resulting in spinal cord compression with or without signs of compressive myelopathy. The craniocervical junction and thoracolumbar spine are the two major locations of spinal cord compression. A medullary syndrome with upper motor neuron signs including gait disturbance, hyper-reflexia, spasticity, clonus, or Babinski sign and muscle weakness are the main clinical features.[25] In Fabry's disease, stroke can be the direct consequence of cardiogenic embolism (caused by hypertrophic cardiomyopathy and conduction abnormalities) or due to a combination of changes in the vessel walls secondary to glycolipid accumulation and abnormal reactivity, and signalling contributing to activation of the coagulation pathway. Patients with Fabry's disease show prominent thickening of the common carotid artery intima, abnormalities of cerebrovascular autoregulation and vasoreactivity, and dysfunction of the cerebral circulation, which seem to have a considerable role in the pathophysiology of neurovascular disease. Lesions might be seen in many areas, including white matter, grey matter, and the posterior circulation.[8]

Barriers to treatment of CNS involvement

The most important drawback of intravenous enzyme replacement therapy is the inability of recombinant enzymes to cross the blood–brain barrier effectively. These therapies have been used to treat lysosomal disorders since the early 1990s,[26] but are only effective for non-neuropathic forms of these disorders. Therefore, obtaining the correct enzyme dose in the brain is a major therapeutic goal. Haemopoietic stem cell transplantation, if performed early enough, might be able to prevent progression of nervous system disease, but procedure-related morbidity and mortality are high, especially for adolescents and adults.[1,27]

Cell therapy

The rationale to treat lysosomal disorders with haemopoietic stem cell transplantation is based on the benefits of cross-correction. This procedure offers the potential of constant secretion of functional lysosomal enzymes that can enter surrounding affected cells by direct receptor-mediated uptake.[28] Transplanted donor cells can also cross the blood–brain barrier and differentiate into microglial cells, thus allowing for improvement of the CNS manifestations. The cellular microenvironment is also modified to allow secretion of cytokines that reduce inflammation, which is known as a major component of the pathophysiology of lysosomal disorders.[29,30]

If full donor chimerism is established, haemopoietic stem cell transplantation can be a single intervention offering a lifelong supply of enzymes. The benefits of the procedure are dose dependent: a higher concentration of enzymes leads to greater benefit. Thus, the use of non-carrier donors is recommended.[31]

Since 1980, a few thousand patients with lysosomal disorders have received this treatment,[31] and different outcomes have been observed depending on the disease type. At present, haemopoietic stem cell transplantation is highly recommended for patients with mucopolysaccharidosis I if they are younger than 2·5 years. It can also be beneficial for patients with metachromatic leukodystrophy and Krabbe's disease if the transplantation is performed shortly after birth;[27,31–35] however, it is currently not indicated for patients with early-onset Tay-Sachs disease and Sandhoff's disease[36] and mucopolysaccharidosis III.[37]

Outcomes for other lysosomal disorders have been variable because of several factors, such as age at diagnosis, age at transplantation (asymptomatic *vs* symptomatic stage), chimerism levels, genotype, type of conditioning (some conditioning drugs are neurotoxic), and graft-versus-host disease.[34] Major risks are still associated with graft-versus-host disease (9% acute grade III–IV and 27% acute grade II).[34,38] However, mortality has been reduced to approximately 10%, with variable rates in different lysosomal disorders.[36]

The beneficial effects of haemopoietic stem cell transplantation on impaired CNS function can be limited by the slow delivery (at least 1 year) of sufficient concentrations of functional enzyme to the CNS.[1,31] In most cases, disease progression is fast, and by the time a sufficient dose of functional enzyme is delivered to the brain via microglial cell repopulation, irreversible damage has already occurred.[31] This limitation could be overcome if a higher dose of enzyme is delivered, and one approach to achieve this aim is by genetically modifying haemopoietic stem cells to overexpress the deficient enzyme. Thus, haemopoietic stem cell transplantation coupled with gene therapy is expected to be more efficacious than is conventional allogeneic transplantation (table 2).[28,31,39] Future studies will determine the safety and efficacy of such a combination approach, but animal models of lysosomal disorders that did not benefit from conventional haemopoietic stem cell transplantation showed effective disease correction when the procedure was coupled with gene therapy.[31] The combination approach has already been tested in patients with metachromatic leukodystrophy in early-phase trials;[40,41] prevention of CNS demyelination and subsequent improvements in peripheral nervous system neuropathy were observed in some patients.

Treatment at an early age is key to improve outcomes because established damage is not easily reversed,

	Allogeneic HSCT	Autologous HSCT plus gene therapy
Donor requirements	Necessity of HLA-matched donor and related delays in the procedure	No need for donors
Risk of transplant-related complications (graft-versus-host disease, infection, and death)	High	Low
Conditioning dose	High	Low
Time for CNS improvement	Long (because of microglial cell repopulation)	Potentially fast if supraphysiological levels of enzyme or vector targeting blood–brain barrier peptides are used
Chimerism *vs* enzyme levels	Limited effect if incomplete donor chimerism	Overexpression overcomes the need for transduction in all cells
Peripheral neuropathy	Limited effect	Might improve if given at asymptomatic or early stages
Remyelination	Limited effect or no improvement	Might induce remyelination if done at asymptomatic or early stages
Immune reconstitution	Long (post-conditioning)	Fast (lower doses of conditioning)
Tissue specificity	Not tissue specific	Can target specific lineages (eg, blood–brain barrier peptides)
Insertional mutagenesis	No risk	Some risk
Toxicity	No toxicity related to enzyme levels	Overexpression might be toxic (eg, Krabbe's disease)
Long-term effects	Stable levels of enzyme with no serious adverse effects; some patients could develop chronic graft-versus-host disease	Unknown

HSCT=haemopoietic stem cell transplantation.

***Table 2*: Benefits and risks of HSCT *vs* HSCT plus gene therapy**

particularly in the CNS. Newborn screening could be followed by autologous haemopoietic stem cell transplantation plus gene therapy using umbilical cord blood to potentially cure the disease. Another alternative treatment approach that might offer hope for treatment of the CNS components of lysosomal disorders is transplantation of neural stem cells.[42] This approach has been shown to be safe in a phase 1 trial of CNS stem cell therapy for infantile or late-infantile neuronal ceroid lipofuscinoses.[43] Further studies in infants (≤1 year in age) will determine the efficacy of this new treatment approach.[43] Haemopoietic stem cell transplantation can also be coupled with intrathecal enzyme replacement therapy to enhance outcomes.[44] Currently, a phase 1 trial with intrathecal delivery of laronidase before and after stem cell transplantation in patients with mucopolysaccharidosis I is underway (ClinicalTrials.gov, NCT00638547).

Enzyme replacement therapy

Enzyme replacement therapy was first conceived as a treatment modality for lysosomal disorders in 1964 by de Duve,[45] who showed that any substance that enters the cell through endocytosis ends up inside the lysosomes. It is now an established treatment for an increasing number of disorders. In many cases, it is the only option for specific treatment. However, the clinically available enzymes cannot cross the blood–brain barrier in adequate amounts because of the absence of mannose-6-phosphate receptors in the endothelial cells of the blood–brain barrier (figure 1), and their effects on the CNS, such as improvement of cerebrovascular disease in Fabry's disease, result from indirect mechanisms.[46] A small proportion of lysosomal enzymes cross the blood–brain barrier, probably through fluid-phase endocytosis.[47] According to findings from animal model studies, therapeutic effects on the CNS might only be achieved when very high doses are administered or during the neonatal period, when the blood–brain barrier expresses mannose-6-phosphate receptors.[48]

A promising strategy to enable enzymes to penetrate the blood–brain barrier is the development of fusion proteins, in which enzyme molecules are attached to peptides or peptidomimetic antibodies that can cross this barrier through receptor-mediated endocytosis, acting as so-called molecular Trojan horses (figure 1). One such approach includes the use of a human insulin receptor monoclonal antibody,[49] which has been tested in rhesus monkeys and is able to deliver sufficient amounts of α-iduronidase,[50] iduronate-2-sulphatase,[51] sulphamidase,[52] and α-*N*-acetylglucosaminidase[53] to the CNS. Other peptides used in fusion proteins include apolipoprotein E, transferrin, and insulin-like growth factor II.[54,55] Fusion proteins are currently being assessed in phase 1 and 2 trials (NCT03053089, NCT02371226, NCT02262338, NCT02597114, and NCT03071341).

Another alternative to reach the CNS is to administer the drug directly into the cerebrospinal fluid (figure 1). This delivery can be done by intrathecal, intracisternal, or intracerebroventricular routes, which can be combined with a simultaneous intravenous treatment for the systemic manifestations. In the cerebrospinal fluid, enzyme molecules can cross the pia–glia limitans interface or reach the brain parenchyma through the perivascular space.[56] Considering the directional flow of the cerebrospinal fluid and the results of experiments in animal models, intracerebroventricular administration might be the most effective of the three routes.[57]

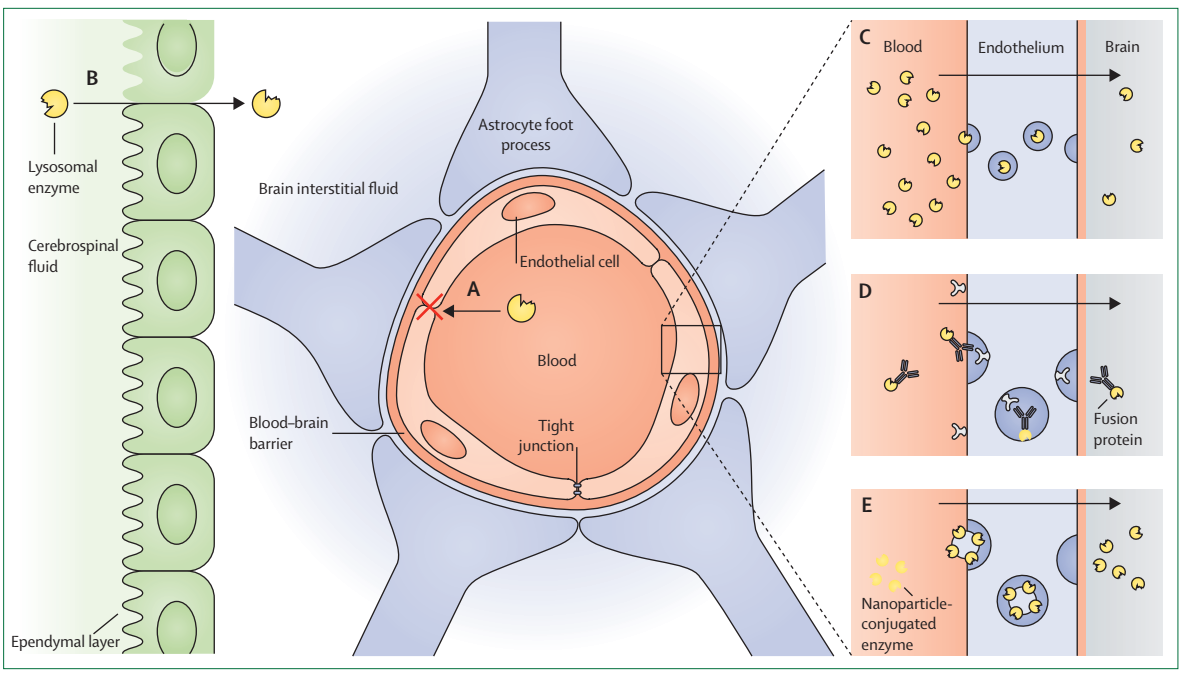


***Figure 1*: Strategies to enable therapeutic enzymes to penetrate or bypass the blood–brain barrier**
Tight junctions prevent the crossing of lysosomal enzymes between endothelial cells (A). Direct infusion into the cerebrospinal fluid through different routes is used to bypass the blood–brain barrier (B). A larger amount of the enzyme might cross the blood–brain barrier when higher doses are used or peripheral clearance is decreased (C). Fusion proteins can cross the blood–brain barrier through receptor-mediated transcytosis (D). Nanoparticles could aid the enzyme to cross the blood–brain barrier through different mechanisms, including transcytosis (E).

Nevertheless, there is also evidence that intrathecal administration could be as effective as intracerebroventricular administration for delivering the enzyme to the deep and superficial brain tissues, while providing higher penetration to the spinal cord.[58] The timing of the treatment seems to be important: late treatment with intrathecal enzyme replacement therapy might not completely reverse substrate storage, whereas early treatment might prevent it.[59] The intrathecal or intracerebroventricular route requires either implantation of devices or repeated lumbar punctures, which can be associated with complications.[60] Inflammatory infiltration in the meninges, which might not be accompanied by signs of meningeal irritation, can also occur.[61] Intrathecal and intracerebroventricular enzyme replacement therapies are being assessed in clinical trials (NCT00638547, NCT01506141, and NCT02055118).[60,62]

An intravenous drug can be optimised to cross the blood–brain barrier even without the use of a molecular Trojan horse. Although a small proportion of the recombinant enzymes can reach the brain, this proportion can be substantially increased by inactivating the terminal sugars of the enzyme, which impairs the peripheral clearance by mannose receptors or mannose-6-phosphate receptors and thus increases the half-life.[63] Moreover, some experimental recombinant enzymes have been shown to cross the blood–brain barrier in higher amounts compared with those currently used in enzyme replacement therapy, which could be due to their hydrophobicity or other unknown properties.[64] Coating the enzyme with nanoparticles can also increase the amount of enzyme delivered to the brain (figure 1).[65]

Gene therapy

Gene therapy is a potential new treatment for lysosomal disorders. The permanent correction of a genetic sequence can be achieved by genetic modification of a specific cell type in the laboratory and its subsequent transplantation (ex-vivo gene therapy) or by direct injection of a vector (in-vivo gene therapy). Both approaches have been used to correct the CNS manifestations in animal models and in early-phase clinical trials involving several lysosomal disorders.

Initial efforts to treat lysosomal disorders before the advent of enzyme replacement therapy relied on allogeneic haemopoietic stem cell transplantation. This approach is applicable in only a limited number of diseases because of several factors, such as the characteristics and progression of the disease, and complications related to allotransplantation. The rationale for ex-vivo gene therapy is to modify autologous haemopoietic stem cells in the laboratory and infuse these cells into patients to repopulate the bone marrow, which addresses the immunological complications arising from allogenic transplantation. It has been tried in several neuropathic lysosomal disorders, but perhaps the most successful example is for metachromatic leukodystrophy. In a phase 1/2 trial, presymptomatic

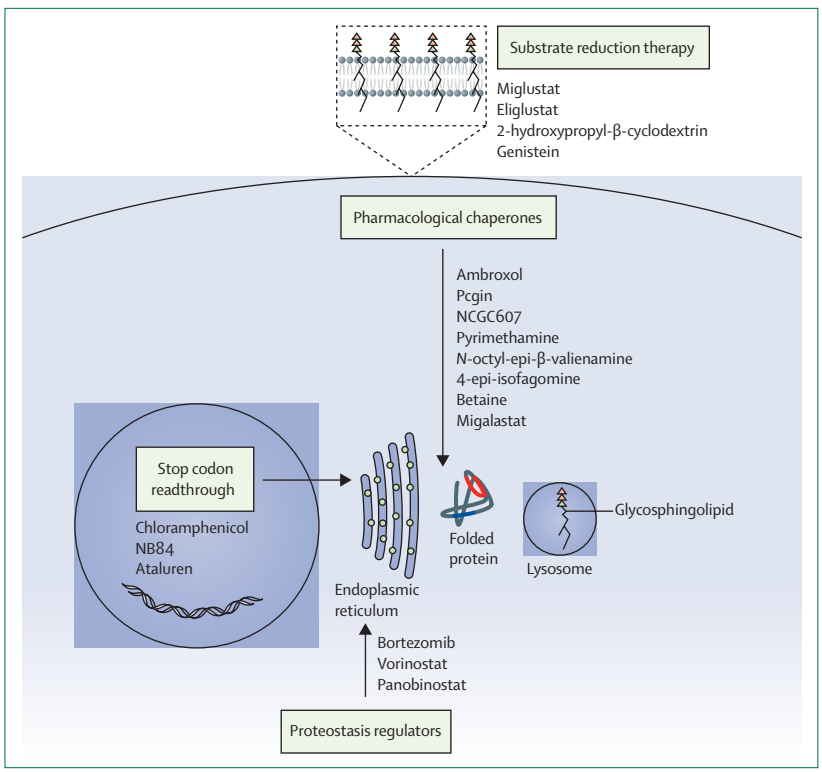


***Figure 2*: Schematic overview of mechanisms of small molecules for the treatment of lysosomal disorders**
Examples of drugs are given for each mechanism.

patients predicted to develop a severe form of disease were treated with autologous haemopoietic stem cells transduced with a lentivirus encoding arylsulphatase A. High expression levels of arylsulphatase A were obtained in both the stem cells and cerebrospinal fluid, and disease progression was stabilised for up to 24 months after treatment.[40] Results from a follow-up study[41] revealed that disease onset was prevented or disease progression was halted in eight of nine patients, providing evidence of the safety and efficacy of this approach. Other preclinical studies have shown that this approach could potentially treat other lysosomal disorders such as Krabbe's disease,[66] mucopolysaccharidosis II (also known as Hunter's disease),[67] and Sanfilippo type A disease.[68]

Another approach to treat the CNS with gene therapy is the direct injection of vectors into the brain parenchyma or cerebrospinal fluid, or the peripheral injection of a vector with high affinity for the brain. This in-vivo injection can circumvent most of the hurdles imposed by haemopoietic stem cell transplantation (namely, the need for myeloablation and a low engraftment success rate), and the use of different adeno-associated virus (AAV) vectors has achieved good results in both preclinical and clinical studies. Preclinical studies were successfully conducted in multiple animal models, including the mouse model of Niemann-Pick disease type C[69] and Tay-Sachs disease,[70] as well as dog models of mucopolysaccharidosis I[71] and Batten disease.[72]

A few trials are under way on the basis of these preclinical results. Most of the clinical studies using this approach aim to correct the phenotype observed in neuronal ceroid lipofuscinoses by intrathecal (NCT02725580) or intracerebral (NCT01414985) routes. Other clinical trials that are ongoing include intravenous injection of an AAV9 vector for mucopolysaccharidosis IIIA (NCT02716246) and intracerebral injections of an AAVrh10 vector in patients with metachromatic leukodystrophy (NCT01801709). A phase 1/2 study[73] has already been done in patients with mucopolysaccharidosis IIIA: four patients were injected with an AAVrh10 directly into the brain, and neuropsychological assessments suggested a moderate improvement in behaviour, attention, and sleep.

On the basis of these results, both ex-vivo and in-vivo gene therapy approaches have the potential to treat the neurological aspect of lysosomal disorders.

Small molecules

In pharmacology, small molecules are low-molecular-weight (usually <700 Da) synthetic organic compounds that have antagonistic or agonistic effects on intracellular and extracellular targets. Examples of small molecules that have been used to treat lysosomal disorders are pharmacological chaperones, proteostasis regulators, substrate synthesis inhibitors, and drugs that increase gene expression by suppressing translation termination (ie, stop codon readthrough; figure 2).

Missense variants can cause protein misfolding, which leads to premature degradation by the endoplasmic reticulum quality control machinery. Pharmacological chaperones can interact with the mutant protein, enhancing its stability and aiding the enzyme to reach the lysosome, where it can have residual function in substrate degradation. Several compounds have been tested as potential pharmacological chaperones for the treatment of neurological symptoms of lysosomal disorders. These compounds include: ambroxol,[74] and most recently pcgin[75] and NCGC607[76] for the treatment of Gaucher's disease; pyrimethamine for late-onset Tay-Sachs disease;[77] *N*-octyl-epi-β-valienamine[78] and, recently, derivatives of 4-epi-isofagomine for β-galactosidase deficiency;[79] betaine for aspartylglucosaminuria;[80] and the only pharmacological chaperone approved in some countries for a lysosomal disorder, migalastat, for the treatment of Fabry's disease.[81]

Protein homoeostasis, or proteostasis, refers to the set of interacting activities that maintain the health of the proteome and the organism, and the proteostasis network consists of chaperones, folding enzymes, and degradation components.[82] Proteostasis regulators could increase lysosymal enzyme function by restoring the function of proteostasis pathways that are disrupted in lysosomal disorders. Several such compounds have shown promise as potential treatments for lysosomal disorders. For example, bortezomib, a proteasome inhibitor approved

for cancer treatment, has been shown to reduce cholesterol accumulation in fibroblasts of patients with Niemann-Pick disease type C[83] and to increase glucocerebrosidase activity in a patient with Gaucher's disease during treatment for multiple myeloma.[84] Besides proteasome inhibitors, histone deacetylase inhibitors, such as vorinostat and panobinostat, have been shown to correct the cholesterol overload in patient-derived Niemann-Pick C1 fibroblasts.[85] Furthermore, the reduction of calcium concentration in the endoplasmic reticulum has been investigated as a potential therapeutic approach, with promising results in Gaucher's disease, Niemann-Pick disease type C, and mucolipidosis type IV.[86] However, the long-term effects of the regulation of the proteostasis cannot be predicted. The link between autophagy impairment and lysosomal disorders has led to novel approaches to modulate cellular clearance, which have been investigated in preclinical studies,[87] but translation to clinical trials is still premature.

The concept of substrate reduction therapy for lysosomal disorders is based on the possibility of a small-molecule drug, usually with oral availability, acting as a functional competitor of the upstream natural compound and partially inhibiting the biosynthesis of the downstream molecules that accumulate within the lysosomes. The first market-approved substrate reduction therapy drug for the treatment of lysosomal disorders was miglustat,[88] a synthetic analogue of D-glucose that inhibits glucosylceramide synthase, for the treatment of Gaucher's disease and Niemann-Pick disease type C. Although miglustat has been shown to reduce the accumulation of glycosphingolipids in mouse models of gangliosidoses, such as Tay-Sachs disease, Sandhoff's disease, and GM1 gangliosidosis, it has not been approved for the treatment of patients because of inefficacy in achieving measurable clinical benefits.[89] Miglustat has been tried in combination with enzyme replacement therapy for neuropathic forms of Gaucher's disease, with no significant benefits;[90] nevertheless, there are anecdotal reports on neurological improvement in some cases.[91] Another substrate inhibitor option for Gaucher's disease, eliglustat, which is a ceramide analogue, was granted marketing authorisation in 2014. By contrast with miglustat, eliglustat can be used as first-line therapy for patients with the non-neuropathic form of Gaucher's disease.[92] However, eliglustat does not cross the blood–brain barrier, limiting its effects on the neuropathic forms of the disease. Preclinical trials with other compounds for the treatment of the neurological symptoms of Gaucher's disease and Fabry's disease are ongoing and have shown promising results.[93,94] Besides miglustat, several cholesterol-lowering agents have been investigated in preclinical and clinical studies of Niemann-Pick disease type C. Intrathecal and intravenous 2-hydroxypropyl-β-cyclodextrin has been shown to have clinical benefits or stabilisation of disease progression in most of the 17 patients reported,[95] although these results still need to be corroborated with data from other studies and clinical trials. Regarding the treatment of neurological symptoms in mucopolysaccharidosis I, II, and III (all subtypes), in-vitro and animal model studies have shown that the administration of genistein, an isoflavone and a protein tyrosine kinase and epidermal growth factor inhibitor, led to a substantial reduction of glycosaminoglycan accumulation.[96] Clinical trials have been done on the basis of these findings and have shown diverse results, although there is no consensus on the optimal dose of the drug.[97]

A nonsense mutation causes a premature stop codon, resulting in the translation of a truncated protein that will be degraded by nonsense-mediated decay. Many lysosomal disorders are caused by this type of loss-of-function mutation and could potentially benefit from drugs that promote the ribosome to read through the premature stop codon. For example, some genetic diseases such as Duchenne muscular dystrophy have market-approved readthrough-inducing drugs and others are in advanced clinical trials.[98] For lysosomal disorders, chloramphenicol has been shown to enhance iduronidase activity in cell lines derived from patients with mucopolysaccharidosis I,[99] and NB84, a designer aminoglycoside, has been shown to reduce glycosaminoglycan accumulation in a mouse model of mucopolysaccharidosis I.[100] More recently, ataluren has been reported to rescue galactocerebrosidase activity in a

Search strategy and selection criteria

We searched MEDLINE for articles published between Jan 1, 1968, and April 1, 2017, with the terms "lysosomal storage diseases" AND "neurologic manifestations"; "hematopoietic stem cell transplantation AND inborn errors of metabolism"; "hematopoietic stem cell transplantation AND lysosomal storage disorders"; "hematopoietic stem cell transplantation AND mucopolysaccharidoses"; "bone marrow transplantation AND inborn errors of metabolism"; "enzyme replacement therapy" AND "central nervous system" AND "lysosomal storage diseases"; "lysosomal storage disorder" AND "gene therapy"; "small molecules AND lysosomal diseases"; "chaperones AND lysosomal diseases"; "proteostasis regulators AND lysosomal diseases"; and "substrate reduction therapy" OR "read-through AND lysosomal disease". We selected publications published after 2000, but we did not exclude commonly referenced and highly regarded older publications. We have only cited articles in English. We also searched reference lists of articles identified by this strategy and selected those we judged to be relevant and free of bias. We included full research articles, case reports, editorials, abstracts published in annals of lysosomal disease conferences, review articles, observational studies, retrospective studies, randomised controlled trials, and book chapters. We also searched for the same terms, alone or in combination, in the ClinicalTrials.gov website. The final reference list was generated on the basis of relevance to the broad scope of this Review.

naturally occurring model of Krabbe's disease,[101] and to increase palmitoyl-protein thioesterase 1 activity in the liver and muscle, but not in any other tissue, of a mouse model of infantile neuronal ceroid lipofuscinoses.[102]

Perspectives and conclusions

Many lysosomal disorders have neurological involvement, making them a difficult target for treatment. However, emerging therapies are overcoming the barriers and providing new hope for affected patients and families. Improved knowledge about the natural history of the neurological manifestations of lysosomal disorders will be important to allow the definition of outcomes for clinical trials, and the identification of the minimally important differences will be paramount for assessment of long-term and short-term efficacies of new therapies for the different lysosomal disorders.

Future newborn screening programmes will enable diagnosis shortly after birth, allowing the initiation of therapy, when indicated, during asymptomatic stages. Newborn screening for mucopolysaccharidosis I and Pompe's disease has already been recommended by the US Federal Committee, and several states in the USA have already incorporated newborn screening for many lysosomal disorders in their regular newborn screening programmes.

The combination of early diagnosis with effective therapies should change the outlook for patients with lysosomal disorders with neurological involvement in the next 5–10 years.

Contributors

All authors contributed equally to this work.

Declaration of interests

RG is on the advisory boards of Actelion, Amicus, BioMarin, Lysogene, Sanofi-Genzyme, Shire, and Ultragenyx, and has received investigator fees, speaker honoraria, or grants to attend scientific meetings from these companies. JAS acts as consultant for and has received a research grant from PTC Therapeutics. The other authors declare no competing interests.

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